example, we will add the suffix “hem” to the string “ant” already in the trie).

Listing 6.5 method insert

```
function insert(node,
s)                                                    #1
  if s == “” then
                                                  #2
    node.keyNode ← true
    return
                                                          #3
  c, tail ←
s.splitAt(0)
   #4
  if node.children[c] != null then
                            #5
    return insert(node.children[c],
tail)                                        #6
  else
    return addNewBranch(node,
s)                                               #7
```

#1 Method `insert` takes a trie node and the string key `s` to be inserted; it returns nothing but has side effects on the trie. Again, we assume that `node` is never `null`: this is a reasonable assumption if this method is implemented as a private method, internally called by the trie’s API `search` method.

#2 Check if the string searched is empty: if it is, since this method is implemented using recursion, we know it has traversed the whole path in the trie.

#3 We are at the target node for the target key: we set current node to a key node to ensure it will store `s`.